

Unearthing InSights into Mars: Unsupervised Source Separation with Limited Data

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Problem statement and contributions

Unsupervised source separation. Retrieving unknown source signals recorded through a mixing operator, with limited prior knowledge about the sources, and only access to a dataset of signal mixtures.

Challenges:

- inherent non-uniqueness
- lacking expert knowledge regarding sources
- limited access to training data (~ 50 data snippets)

Our contribution. An unsupervised source separation method applicable to domains with limited access to data, e.g., when dealing with extraterrestrial data.

Example: Marsquake cleaning

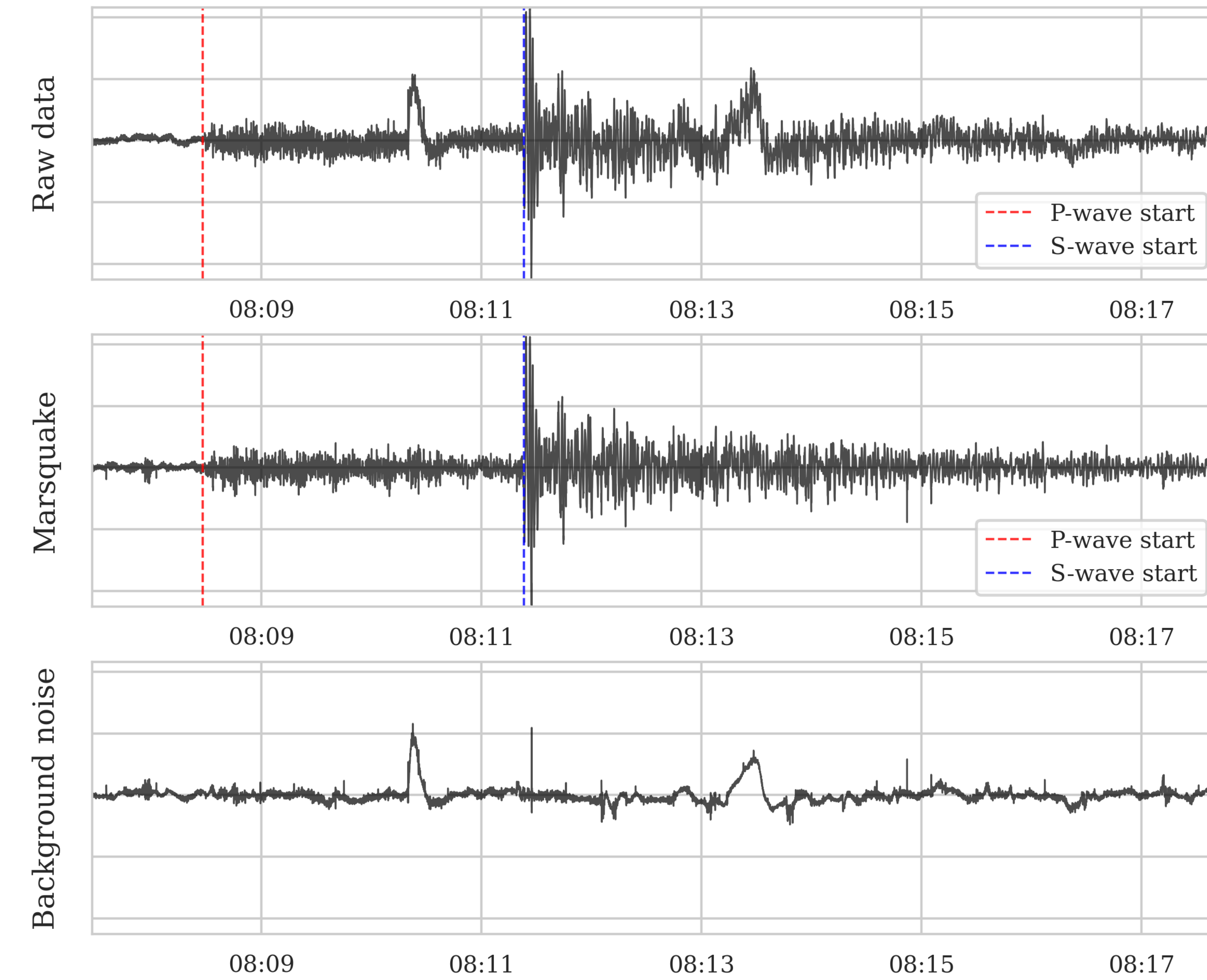


Figure 1: Unsupervised separation of background noise, including thermally induced microtilts (glitches), from a marsquake recorded during the NASA InSight mission.

Wavelet Scattering Covariances

A quasi-block diagonal approximation to second-order moments of wavelet scattering network output $\Phi(\mathbf{x}) := \mathbb{E} [S(\mathbf{x}) S(\mathbf{x})^\top]$

- $S(\mathbf{x}) := [\mathbf{W}\mathbf{x}, \mathbf{W}|\mathbf{W}\mathbf{x}|]^\top$ and $\mathbf{W}$ the wavelet transform
- expectation over time (estimated by average pooling)
- a low-dimensional representation of stochastic processes
- capable of capturing non-Gaussian properties
- scalable as block matrices in $S(\mathbf{x}) S(\mathbf{x})^\top$ are quasi-diagonal

Unsupervised Source Separation

Objective. Retrieve unknown source $\mathbf{s}(t)$ from $\mathbf{x}(t) = \mathbf{s}(t) + \mathbf{n}(t)$ using observed samples $\{\mathbf{n}_k(t)\}_{k=1}^K$ as prior information.

Proposed method. Cast source separation as an optimization problem in the wavelet scattering covariance representation space

$$\hat{\mathbf{s}} := \arg \min_{\mathbf{s}} \left[\mathcal{L}_{\text{data}}(\mathbf{s}) + \mathcal{L}_{\text{prior}}(\mathbf{s}) + \mathcal{L}_{\text{cross}}(\mathbf{s}) \right]$$

$$\text{data misfit loss: } \mathcal{L}_{\text{data}}(\mathbf{s}) = \frac{1}{K} \sum_{k=1}^K \frac{\|\Phi(\mathbf{s} + \mathbf{n}_k) - \Phi(\mathbf{x})\|_2^2}{\sigma^2(\Phi(\mathbf{x} + \mathbf{n}_k))},$$

$$\text{prior loss: } \mathcal{L}_{\text{prior}}(\mathbf{s}) = \frac{1}{K} \sum_{k=1}^K \frac{\|\Phi(\mathbf{x} - \mathbf{s}) - \Phi(\mathbf{n}_k)\|_2^2}{\sigma^2(\Phi(\mathbf{n}_k))},$$

$$\text{cross-covariance loss: } \mathcal{L}_{\text{cross}}(\mathbf{s}) = \frac{1}{K} \sum_{k=1}^K \frac{\|\Phi(\mathbf{s}, \mathbf{n}_k)\|_2^2}{\sigma^2(\Phi(\mathbf{x}, \mathbf{n}_k))}$$

Conclusion

We reaped the benefits of the inductive bias built into scattering covariances, enabling us to obtain low-dimensional representations that characterize a wide range of non-Gaussian properties of stochastic processes without pretraining, leading to a data frugal source separation method.

Probabilistic recovery guarantee

Theorem 1. Let $\mathbf{x} = \mathbf{s} + \mathbf{n}$ with $\mathbf{s}$ and $\mathbf{n}$ two independent processes. Let us assume we have two processes $\hat{\mathbf{s}}$ and $\hat{\mathbf{n}}$ with $\mathbf{x} = \hat{\mathbf{s}} + \hat{\mathbf{n}}$ such that:

- (i) $\mathbf{n}$ has a maximum entropy distribution under moment constraints $\mathbb{E}\{\Phi(\mathbf{n})\}$
- (ii) $\hat{\mathbf{n}}$ has a maximum entropy distribution under moment constraints $\mathbb{E}\{\Phi(\hat{\mathbf{n}})\}$
- (iii) $\mathbb{E}\{\Phi(\hat{\mathbf{n}})\} = \mathbb{E}\{\Phi(\mathbf{n})\}$
- (iv) $\hat{\mathbf{s}}$ and $\hat{\mathbf{n}}$ are independent
- (v) The Fourier transform $\hat{p}_{\mathbf{n}}$ of the distribution $p_{\mathbf{n}}$ of $\mathbf{n}$ is non-zero everywhere.

$\Rightarrow \mathbf{n} \stackrel{d}{=} \hat{\mathbf{n}}$ and $\mathbf{s} \stackrel{d}{=} \hat{\mathbf{s}}$ where the equality is on the distribution of the processes.

Stylized example: Turbulent signal separation

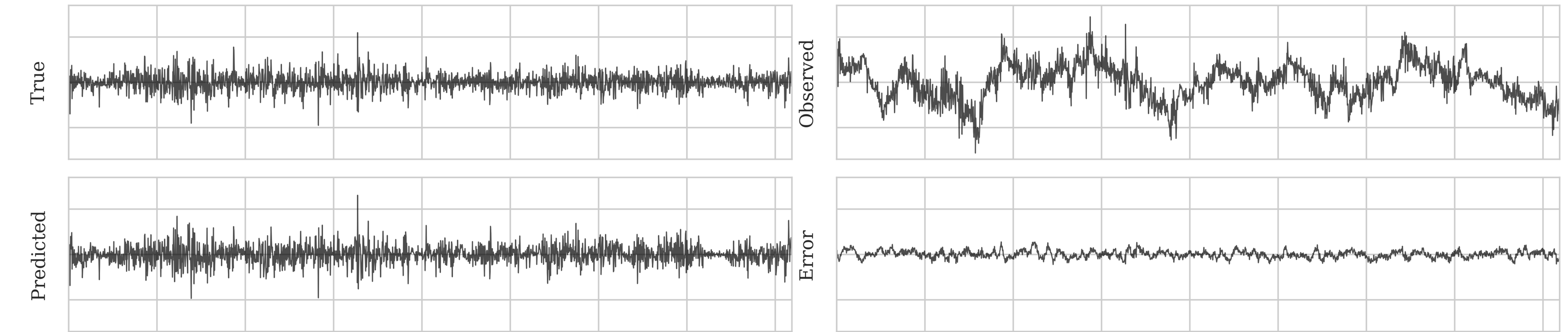


Figure 2: Unsupervised source separation applied to the multifractal random walk data with a turbulent additive signal. The vertical axis is the same for all the plots.

Separating glitches: Seismic data from Mars

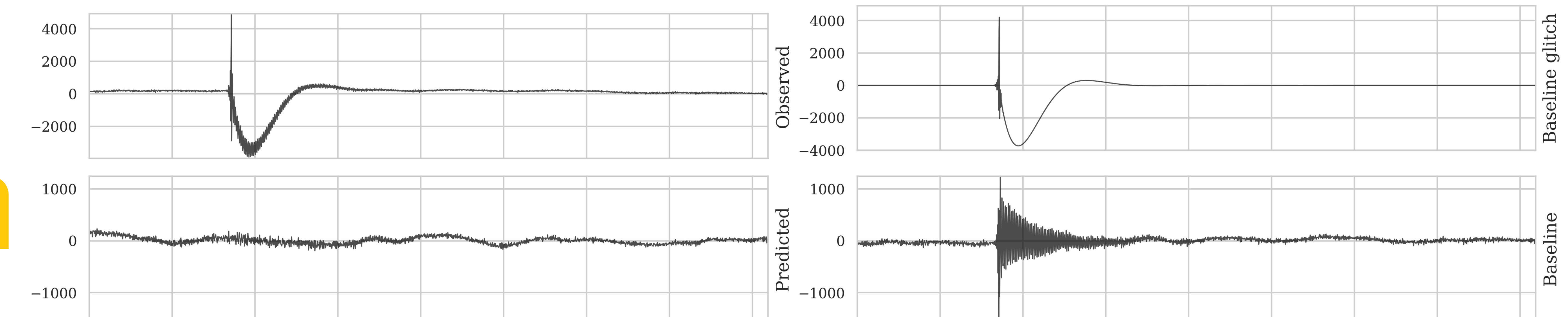


Figure 3: Unsupervised source separation for glitch removal. Our approach successfully separates the glitch using only glitch-free data snippets, while the baseline approach fails to accurately characterize the glitch, resulting in significant residual oscillations.